

Computational classification of SOD2 protein in coral and symbiotic algae from public genomic and transcriptomic data

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ABSTRACT

SOD2, or Manganese/Iron Superoxide Dismutase, plays a crucial role in maintaining cellular health across the tree of life. Existing literature is vague in identifying types of SOD2 proteins expressed in both corals and their symbiotic algae; most agree on the presence of a) mitochondrial targeting MnSOD in corals, and b) likely Mn dependent SOD2 in algae. This paper presents evidence for four classes of SOD2 proteins between coral hosts and their symbiotic algae. Public genomic and transcriptomic data reveal mitochondrial targeting and secreted versions of SOD2 among a wide range of cnidaria species, and chloroplast targeting (with a bipartite leader architecture) and secreted versions of SOD2 among several symbiodinium species. The mature/catalytic region of SOD2 is well conserved within each class, but highly differentiated between classes. This paper presents the mature region HMM profiles for each class, with threshold scores high enough such that they are predictive of leader type.

Introduction

SOD2 proteins protect cells against oxidative damage by converting toxic superoxide radicals into hydrogen peroxide and diatomic oxygen. SOD2 proteins can bind either manganese (MnSOD) or iron (FeSOD) as co-factors; their structure consists of conserved residues His26, His74, Asp159, and His163 (human sequence coordinates) forming the anchors of a trigonal bipyramidal metal pocket and variations in the outer shell, based on bound metal, to ensure superoxide dismutase activity [1]. SOD2 is present throughout tree of life, with bacterial SOD2 often preferring Fe as cofactor, mammalian SOD2 almost always requires Mn as a cofactor to function in mitochondria, and metal preference not always phylogenetically constrained [2].

Existing literature is vague on the types of SOD2 proteins in coral hosts and their symbiotic algae. While there is consensus on the presence of mitochondrial MnSOD in corals, other forms of SOD2 in coral and symbiotic algae are rarely discussed/analyzed separately. [3] reports experimental evidence for secreted superoxide dismutase activity in *Stylophora pistillata* without identifying the source of such activity. [4] experimentally identifies two forms of MnSOD in *Hydra vulgaris*, a non-coral species in the cnidaria phylum. [5] computationally compares various annotated MnSOD and FeSOD sequences from various symbiodinium species, with grouping of sequences, based on putative annotation, inconsistent with findings of this paper. Indeed, most transcriptomic analyses in the field depend on blasting against UniProt or NCBI, or matching against the HMM profile K04564, a pan-metagenomic KEGG ortholog ([6]) for SOD2. Neither approaches provide resolution beyond putative superoxide dismutase activity.

This paper presents results from an unbiased computational search and grouping of SOD2 protein sequences, using NCBI deposited genome sequences from 36 coral and 10 symbiodinium species, as well as RNAseq datasets from 5 papers. The goal of this work is to present specific and unambiguous sequence-level profiles for SOD2 proteins in *Anthozoa* and *Dinophyceae*. Although these protein groups have features that are suggestive of biological and biochemical mechanism, the scope of this paper is limited to computational;

there is no experimental work, beyond referenced RNAseq experiments, to functionally validate the different classes of proteins, determine their metal preference, or likely cellular location.

Specifically, a protein sequence database was constructed from a) protein sequences submitted to and associated with NCBI genome accessions, b) protein sequences exhaustively detected using 25k+ KO HMM profiles ([6]) on 6-frame translations of genomic sequences submitted to and associated with NCBI genome accessions, and c) protein sequences generated from transcriptomes from 5 published RNAseq studies. From this database, sequences matching the SOD2 K04564 profile, having PF00081 and PF02777 Pfam ([7]) FeSOD N and C-terminal domains, and having four key metal binding pocket residues ([1]) were retained.

A leader discovery algorithm was employed to look for, in predicted exons if protein sequence was obtained through a gene model (above (a)) or in upstream 3-frame translations (above (b)) or in the protein sequence (above (c)), possible start codons upstream of the PF00081 match. For each sequence retained from the original protein sequence database, zero (if a start codon cannot be found) or more candidate K04564 sequences may be generated from this leader discovery algorithm. This algorithm is described in more detail in the methods section below.

DeepLoc 2.1 ([8]) classification of candidate K04564 sequences and MUSCLE multi-sequence alignments ([9]) were used to group protein sequences, for *Anthozoa* and *Dinophyceae* separately, by leader type, localization, and sequence similarity.

Results

Four classes of SOD2 proteins, termed mt_cnidaria, sp_cnidaria, sp_ctp_algae, and sp_algae, are evident from the candidate sequences. Figure 1 shows a breakdown of the number of proteins per class, by taxonomy. Transcriptomic data are grouped in the first columns on the left.

Talk about transcriptomic data.

Talk about mature sequence.

Figure 2

IQtree of mature enzyme sequence (3 AA of PF00081 to 100 AA of PF02777) forms 4 clades, with sp_cnidaria being paraphyletic, the rest are monophyletic, hard to assign metals from this as sequence similarity may not correspond to metal preference. This establishes that

A mitochondrial MnSOD exists across all corals

In coral hosts, the mitochondrial version of SOD2 includes of a mitochondrial transit peptide (mTP) XXX amino acids upstream of N-terminal PF00081 domain, and the secreted version includes a signal peptide (SP) XXX amino acids upstream of PF00081. These two classes are termed mt_cnidaria and sp_cnidaria, respectively.

High mature sequence similarity. HMM scores. Very likely Mn.

What about Mt? Likely different across species? Or also conserved?

Evidence of a secreted SOD2, with unknown metal preference, shown in the genome and transcriptome of some coral species

Paraphyletic, evolutionary lineage tree forms clusters by type: sea anemone, stony, soft.

Transcript counts from one RNAseq experiment

Missing from acropora genomes, which explains why a. tenuis transcripts does not have such a sequence.

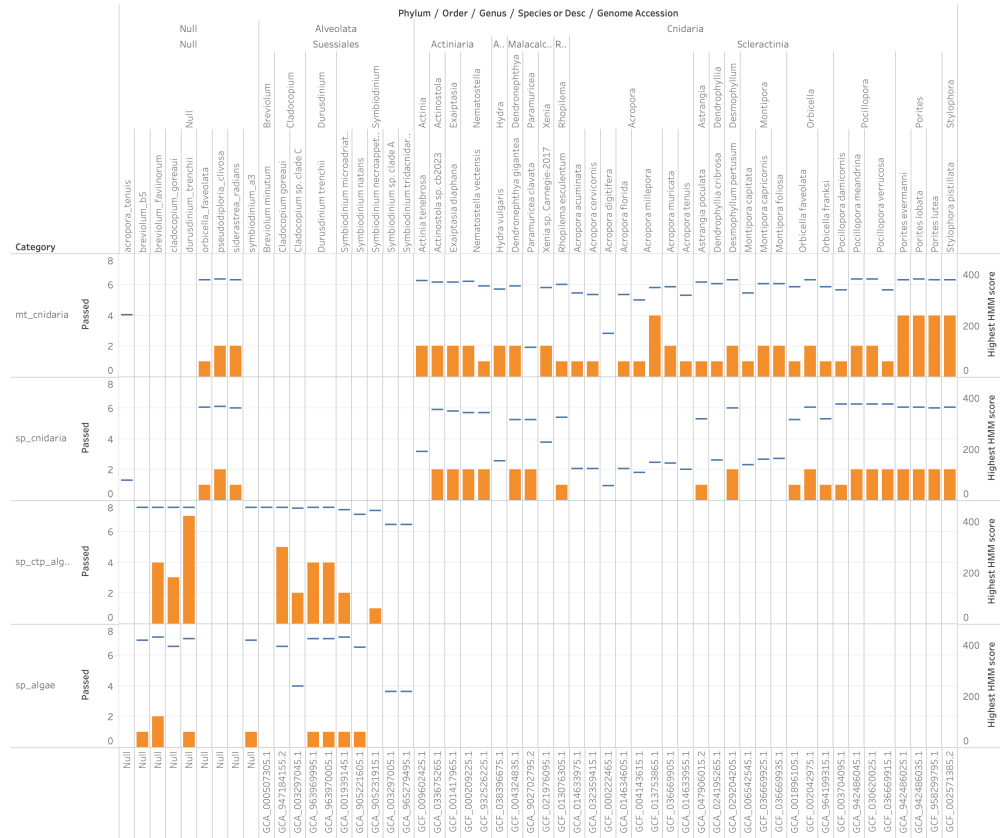


Figure 1. Sample figure caption.

Just one study. But uncertain if it's linked to this.

A likely chloroplast targeting SOD2 protein, with a bipartite leader, is conserved across multiple species of symbiodinium

In algae, the chloroplast targeting version of SOD2 includes a bipartite leader. DeepLoc 2.1 consistently finds a SP leader XXX amino acids upstream of PF00081, and also consistently categorizes the SP-cleaved copies of the proteins as localized to plastid. DeepLoc employs different mechanisms (or "heads" in their ML architecture) for classifying localization and finding leader sequences, and was not trained on symbiodiniums. The SP cleavage point sits XXX amino acids upstream of PF00081, suggesting a bipartite leader architecture of XXX AAs of SP, YYY AAs of cTP, then PF00081. This class is termed *sp_ctp_algae*, in figures and tables in this paper.

Separately, a secreted version of SOD2 with only a SP, XXX amino acids upstream of PF00081, also exist in algae. This class is termed *sp_algae*.

A fourth class of SOD2, with distinct mature protein sequence, may exist in symbiodinium, with weak expression evidence

May be secreted? Transcript counts from one RNAseq experiment suggest very weak if any expression.



Figure 2. Sample figure caption.

The mature protein sequences of the four classes are cleanly distinguishable from each other and covers almost all of detectable K04564 sequences

The mature enzyme sequence, w/o leader, form distinctive clades that do not overlap. Furthermore the HMM threshold scores for mature enzyme group profiles show a large drop with F=1 (i.e. no FPs or FNs) at very clear boundaries. Mature enzyme sequence correlates with leader, and appear to also with target compartment

Generated 202 K04564 matches from our genomic and RNase datasets, reduced down to 114 unique mature enzyme sequences that possess all key residues and spans from 3rd AA of PF00081 to 100 AA of PF02777, so removing any partial matches due to bad genome assemblies or truncated gene models.

this is all_k04564_matches_mature_region.faa and the .tsv file

12 out of 144 fell below the above mature profile thresholds, rest belong to one of the 4 clades

4 are similar to or is CAE7459428.1 and is just below the threshold for sp_ctp_algae; CAE7459428.1 would have been a hit for sp_ctp_algae if its SP signal was stronger.

8/144, or 6

Final HMM profile for each group, with threshold scores for full protein and probably mature proteins, and rules

c. goreau should have had one, but in CAL1167723.1, PF00081 did not match well, if we define PF00081 from position 11 not position 3, may have been ok

Mature HMM vs leader

Should be SP-cPT? DeepLoc reports Golgi as targets for both but do not report SP for them, SignalP reports weak SP, but sequence kind of matches BGNK01003376.1_K04564_a3776c6c u60 OZ257276.1_K04564_ace37290 u60

Should be SP? These are the same sequence. DeepLoc thinks they are not SP and for plastid. Mature sequence definitely matches better for SP_algae but is below threshold. CAL1167723.1_with_leader_u21_PF00081_M evm.model.scf7180000356577.1_with_leader_u21_PF00081_M Issue is that we are missing 11 bps of PF00081

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